NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2 QUESTIONS AND SOLUTIONS.

SECTION A : 5 marks each

1. Simplify $\dfrac{10^{2025}-10^{2026}}{10^{2024}-10^{2025}}$ to its simplest form.

   Solution

   $$\frac{10^{2025}-10^{2026}}{10^{2024}-10^{2025}}=\frac{10^{2024}(10-10^2)}{10^{2024}(1-10)}=\frac{10-100}{-9}=10.$$

2. Dorris is driving on a highway. She notices the odometer reading, 24942 km, is a palindromic number, meaning that it is not changed when reversed. "Hmm," she thinks, "it should be a long time before I see that again". However, it only takes one hour for the odometer to once again show a palindromic number! How fast is Dorris driving?

   Solution

   The next palindromic number after 24942 is 25052. Therefore Dorris has gone $25052-24942=110$ km in one hour. Her speed is 110 km/hr.

3. Six girls are holding placards written on with letters A, B, C, D, E, F. We can call the girls by their placard letter. The girls make 2 rows in such a way that girl E is at the position shown below. Girl A is to the left of F and D is directly below F. Indicate on the figure the location of the rest of the 5 girls.

		E

   Solution

A	F	E
C	D	B

   Note: B and C can be swapped/interchanged.

4. If $6x+5y=4$, find the value of $64^x\times 32^y$.

   Solution

   $$64^x\times 32^y=2^{6x}\times 2^{5y}=2^{6x+5y}=2^4=16.$$

5. Pamella plays basketball for her school. Her free throw shooting percentage for the season was 76% exactly before today. During tonight's game she scores all five free throws, bringing her percentage up to 80%. How many free throws has Pamella scored for the entire season (including tonight)?

   Solution

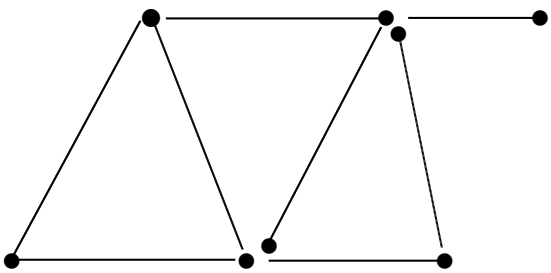
   Let us call her shots taken before tonight y. Her shots made can be represented as $0.76y+5=0.80(y+5)$ where the left side is shots made before tonight +5, while the right side is 80% of all shots made. Solving this equation for y gives $y=25$. Therefore the left side (free throws for the entire season including tonight) becomes $0.76\times 25+5=19+5=24$.

6. Last year 2024, Mariam primary School sent a team of 5 pupils to the National Mathematics Contest. In this year 2025, the school has sent the same team except the oldest member (completed primary seven) has been replaced with a younger pupil. If the average team member age has remained the same for both years, how many years younger is the new member than the replaced one?

   Solution

   Let a_1, a_2, a_3, a_4 be the ages of four pupils who have sat the two contests, a_{old} be the age of the pupil who completed primary seven and a_{new} be the age of the new pupil added to the team. Then $\frac{a_1+a_2+a_3+a_4+a_{old}}{5}=\frac{a_1+a_2+a_3+a_4+4+a_{new}}{5}$. We obtain $a_{old}=4+a_{new}$. This shows that the new pupil is 4 years younger than the old pupil who completed primary seven.

7. A special matchbox contains 10 matchsticks. Maria uses the matchsticks to make patterns of triangles as shown in the figure below. How many matchsticks will she need to make 11 such triangles?

   Solution

   Observe a pattern $\frac{1\triangle}{3}, \frac{2\triangle}{5}, \frac{3\triangle}{7}, \frac{4\triangle}{9}, ..., \frac{11\triangle}{?}$ After solving, $?=23$. Alternatively, let x be the number of triangles and y be the number of sticks, then $y=2x+1$, when $x=11$, $y=23$. **Note:** Someone can draw the 11 triangles and counts the sticks.

8. The price of a bus ticket was increased by 5%. Then, in a special offer to their customers, the price was reduced by 20%. It was now shs 4000 less expensive than its original price. What was the original price of the bus ticket?

   Solution

   Let the initial price be x. After the increase the price became $1.05x$. After the special offer the new price became $1.05x\times 0.8=0.84x$. Since it was now shs 4000 less expensive than its original price, we have $x-0.84x=4000$

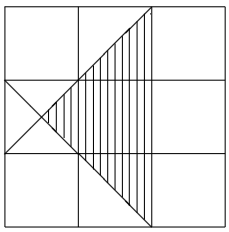
implying $0.16x = 4000$. Solving the equation for x gives $x = 25000$. The original price of the bus ticket is Shs.25, 000.

9. Nathan eats three equal-sized bowls of porridge, one after the other. When he has eaten $\frac{3}{7}$ of the total amount of porridge, what fraction of the porridge in the second bowl has he eaten?

Solution

Let x be the fraction of the second bowl that Nathan has eaten. Each bowl makes up one-third of the porridge. Therefore the fraction of all the porridge that Nathan has eaten is $\frac{1}{3} + \frac{1}{3}x$. Therefore $\frac{1}{3} + \frac{1}{3}x = \frac{3}{7}$ We multiply both sides of this equation by 21 to obtain $7 + 7x = 9$. Hence $7x = 2$ so, $x = \frac{2}{7}$.

10. In the diagram, a square is divided into 9 equal small squares. What fraction of the big square is shaded?



Solution

Observe that the shaded small squares are one full, two 1/2 way shaded and one 1/4 way shaded. It follows that $\frac{2\ 1/4}{9}$ which reduces to $\frac{9}{4} \times \frac{1}{9} = \frac{1}{4}$

SECTION B : (10 marks each)

11. In a school, one fifth of the students have blue eyes. One tenth of the left-handed students have blue eyes. One quarter of the right-handed students have blue eyes. What fraction of the students are left-handed?

Solution

Suppose that l is the number of left-handed students and r is the number of right-handed students. Then the total number of students is $l + r$, and the fraction that are left-handed is $\frac{l}{l+r}$. There are $\frac{1}{10}l$ left-handed students with blue eyes and $\frac{1}{4}r$ right-handed students with blue eyes. Since one fifth of all the students have blue eyes, we have $\frac{1}{10}l + \frac{1}{4}r = \frac{1}{5}(l + r)$. By multiplying both sides of this equation by 20, we obtain $2l + 5r = 4(l + r)$. That is, $2l + 5r = 4l + 4r$. Therefore $r = 2l$. Therefore the fraction of the students who are left-handed is given by $\frac{l}{l+r} = \frac{l}{l+2l} = \frac{l}{3l} = \frac{1}{3}$.

Alternatively, Total number $= l + r = 1 + 2l = 3l$.
It follows that $l = \frac{\text{total number}}{3} = \frac{1}{3}$ of total number.

12. Anna asked Betty "How old are your parents?". Betty answered: "My mother is 4 years older than my father. When the sum of my parents' ages was 84, my age and that of my two sisters were 14, 10, and 6, respectively. Currently, the sum of my parents' ages is twice the sum of my age and that of my two sisters." Work out the current ages of all family members.

Solution

Suppose that n years ago is when the sum of Betty's parents' ages was 84. Then now, the sum of Betty's parents' ages is $84 + 2n$, and the sum of the ages of Betty and her sisters is

$$(14 + n) + (10 + n) + (6 + n) = 30 + 3n.$$

Thus $84 + 2n = 2(30 + 3n)$, so $42 + n = 30 + 3n$, and so $n = 6$. Therefore, the sum of Betty's parents' ages is $84 + 2 \times 6 = 96$. Denote by a and b the ages of Betty's mother and father. Then $a + b = 96$ and $a - b = 4$. Simplifying gives $2a = 100$.

Betty's mother is $a = 50$ years old and her father is $b = 46$ years old. Betty is 20 and her younger sisters are 16 and 12.

13. A data set consists of 6 (not distinct) positive integers: $1, 7, 5, 2, 5$, and X. The average (arithmetic mean) of the 6 numbers equals a value in the data set. What is the sum of all possible values of X ?

Solution

First note that the mean of the elements of the data set excluding X is 4. The given condition will therefore be met if $X = 4$, and if $X \neq 4$, then the mean will not be X. Because X is a positive integer, the mean of the full data set must be greater than or equal to $21 \div 6 = 3.5$. The only other possibilities for the mean are therefore 5 and 7. The mean will be 5 if and only if $1 + 7 + 5 + 2 + 5 + X = 6 \times 5$, which implies that $X = 30 - 20 = 10$. The mean will be 7 if and only if $1 + 7 + 5 + 2 + 5 + X = 6 \times 7$, which implies that $X = 42 - 20 = 22$. The sum of the possible values of X is $4 + 10 + 22 = 36$.

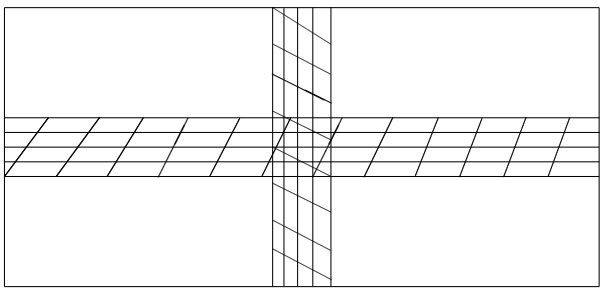
Alternatively,
Sum= $20 + X$ and average= $\frac{20+X}{6}$. Since the average has to be an integer, then $x = 4 (\text{mod } 6)$. When $X = 4$, average= 4 which satisfies the given condition that the average must be in the data set. If $X > 4$, $X \neq$ average considering all other numbers would be less than X.
Therefore, average can only be 5 and 7.
For average = 5, $20 + X = 30$ which results in $X = 10$
For average = 7, $20 + X = 42$ which results in $X = 22$
Sum of vlues of X is $4 + 10 + 22 = 36$

14. Find all pairs of positive integers whose sum is 133 and highest common factor (HCF) is 19. Show your working.

Solution

Assume the pair of numbers to be x and y. Since $\text{HCF}(x, y) = 19$, it implies that $x = 19p$ while $y = 19q$ where p and q are positive integers. Using the given information, we obtain $x + y = 133$ so, $19p + 19q = 133$ implying $p + q = 7$. The possible values of the pair (p, q) are $(1, 6), (2, 5), (3, 4)$. Therefore the possible values of (x, y) are $(19, 114), (38, 95), (57, 76)$.

15. Two roads, each of width 5m and crossing at right angles, have been made in the middle of a rectangular garden measuring 50m by 35m as shown in the diagram below. What area is not covered by the roads?



Solution

The area of the rectangular garden $= 50 \times 35 = 1750\ m^2$. Area covered by the roads $= (50 \times 5) + (35 \times 5) - (5 \times 5) = 400\ m^2$. We subtract to remove the crossroads portion of the roads that is counted twice. Thus the area not covered by the roads $= 1750 - 400 = 1350\ m^2$.

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